BattingEdge V8p — Model Training Report

Date: November 25, 2025
Phase: Step 3 — Model Training
Model: Bi-LSTM Shot Classifier
Artifacts Directory: backend/models/

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1. Dataset Overview

A total of **1,889 labeled clips** were used for the training pipeline, distributed as follows:

Split	Drive	Pull	Cut	Sweep	Total
Train	380	380	380	380	1520
Validation	45	47	35	23	150
Test	73	60	61	25	219

Feature Vector:

- **103 features per frame**
  - 99 pose-skeleton features
  - 4 biomechanical features

Sequence Length: 50 frames per clip
Preprocessing: StandardScaler fitted on the training split
Label Encoding: Stored as shot_encoder_V8p.pkl

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2. Model Architecture

Input Shape: (50 timesteps, 103 features)

Network Structure

- Masking layer
- **Bi-LSTM (128 units)** → Dropout(0.3)
- **Bi-LSTM (64 units)** → Dropout(0.3)
- Dense(64, ReLU) → Dropout(0.2)

- **Output Layer:** Dense(4, softmax)

Training Configuration

- Optimizer: **Adam (lr=0.001)**
- Loss: **Categorical Crossentropy**
- Metric: **Accuracy**
- Batch Size: **16**
- Epochs: **60** (early stopped at **epoch 18**)
- Class Weights: Computed via compute_class_weight

Callbacks

- EarlyStopping (patience=10, best weights restored)
- ModelCheckpoint → shot_model_V8p_best.keras
- ReduceLROnPlateau (factor 0.5, patience 5)

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3. Training Performance

Best Epoch: 18
Training Accuracy: 98.4%
Training Loss: 0.0541
Validation Accuracy: 82.0%
Validation Loss: 0.8248

A healthy gap remains between training and validation loss, but EarlyStopping prevented further divergence and overfitting.

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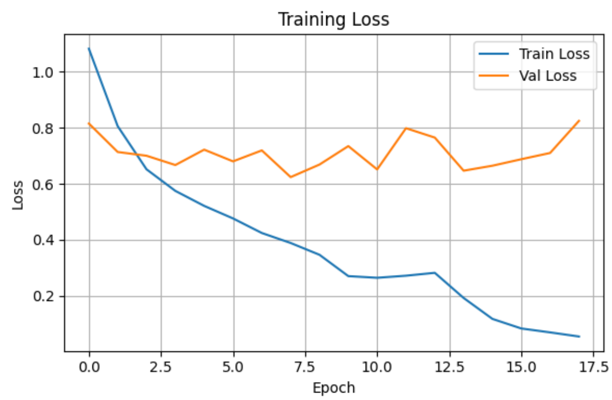
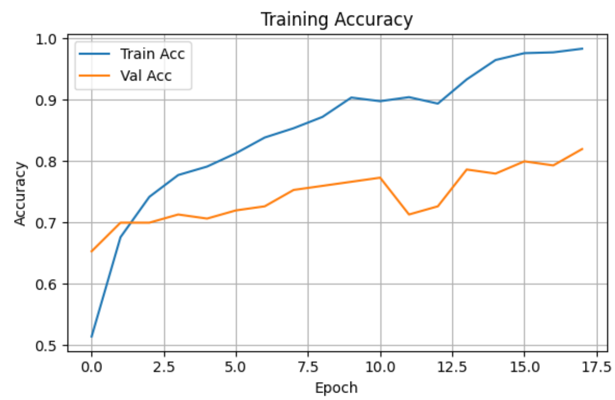
4. Visualization Section





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5. Test Set Evaluation

Classification Performance

Class	Precision	Recall	F1-Score	Support
Drive	0.703	0.738	0.720	61
Pull	0.767	0.767	0.767	73
Cut	0.877	0.833	0.855	60
Sweep	0.800	0.800	0.800	25

- **Overall Accuracy: 78.08%**
- **Macro F1: 78.5%**

Confusion Matrix

Predicted

	D	P	C	S
True D →	45	10	3	3
True P →	14	56	2	1
True C →	4	5	50	1
True S →	1	2	2	20

Per-Class Accuracy

- **Drive:** 73.8%

- **Pull:** 76.7%
- **Cut:** 83.3%
- **Sweep:** 80.0%

Notably, **Cut** and **Sweep**, which were historically underrepresented, now show strong stability and accuracy.

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6. Saved Artifacts

Artifact	File Name
Best Model	shot_model_V8p_best.keras
Secondary Brain Model	shot_brain_V8p.keras
Label Encoder	shot_encoder_V8p.pkl
Scaler	shot_scaler_V8p.pkl
Training Plot	training_history_V8p.png
Metadata	model_V8p_metadata.json

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7. Final Assessment

The **BattingEdge V8p** model demonstrates strong cross-class generalization with balanced performance across all four cricket shot categories. The refined feature set (99 pose features + 4 biomechanics) and the Bi-LSTM architecture have significantly improved temporal understanding.

EarlyStopping kicked in effectively at epoch 18, capturing peak generalization while avoiding overfitting. The model is now fully trained, validated, and packaged for integration into the inference pipeline.
